



SESSION 18

Systems of Linear Equations & Rank

Systems $Ax = b$ · Matrix Rank · Gaussian Elimination · Null Space & Column Space

Systems of Linear Equations ($Ax = b$)

Row Echelon Form & Gaussian Elimination

Rank of a Matrix

Existence & Uniqueness of Solutions

The Four Fundamental Subspaces

Column Space (Range)

Null Space (Kernel)

Inverse Problems in ML

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)



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01 · Systems of Linear Equations ($\mathbf{Ax} = \mathbf{b}$)

A system of linear equations can be written as $\mathbf{Ax} = \mathbf{b}$, where $\mathbf{A}$ contains coefficients, $\mathbf{x}$ represents unknowns, and $\mathbf{b}$ contains target values.

02 · Row Echelon Form & Gaussian Elimination

Gaussian elimination is an algorithm that uses elementary row operations to transform a matrix into **Row Echelon Form (REF)**, allowing us to solve systems of equations systematically via back-substitution.

03 · Rank of a Matrix

The **rank** of a matrix, $\text{rank}(A)$, is the maximum number of linearly independent rows or columns. It measures the dimension of the output space of the transformation.

04 · Existence & Uniqueness of Solutions

For $A\mathbf{x} = \mathbf{b}$ (where A is $m \times n$):

- **No solution:** If $\text{rank}(A) < \text{rank}([A|\mathbf{b}])$ (inconsistent system).
- **Unique solution:** If $\text{rank}(A) = n$ (system is full rank and square).
- **Infinite solutions:** If $\text{rank}(A) < n$ (underdetermined system, free variables exist).

05 • The Four Fundamental Subspaces

Every linear operator has four key subspaces: Column Space, Null Space, Row Space, and Left Null Space. Their dimensions are linked by the Rank-Nullity Theorem.

RANK-NULLITY THEOREM

$$\text{rank}(A) + \text{nullity}(A) = n$$

06 • Column Space (Range)

The column space, $\text{Col}(A)$, is the span of the columns of A . It contains all vectors $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has a solution.

07 • Null Space (Kernel)

The null space, $\text{Null}(A)$, is the set of all vectors $\mathbf{x}$ that map to the zero vector: $A\mathbf{x} = \mathbf{0}$.

08 • Inverse Problems in ML

When solving linear regression, we solve the normal equation $A^T A \mathbf{x} = A^T \mathbf{b}$. If $A^T A$ is not full rank, it has no inverse, requiring **L2 regularization** (Ridge regression) to make it full rank.

09 · Quick Reference Card



Systems & Rank Cheat Sheet

- **Rank:** Maximum independent rows/columns.
- **Inconsistent:** Equations contradict, no solution.
- **Full Rank:** Rank equals $\min(m, n)$.